

Sajeevan (Saj) Veeriah

MECHATRONICS & ROBOTICS ENGINEER | AI/ML | EMBEDDED SYSTEMS | END-TO-END AUTOMATION

Geelong, VIC | Open to Melbourne and broader Victoria | +61 498 586 654 | sajeevanveeriah@gmail.com

linkedin.com/in/sajeevan-veeriah | sajeevanveeriah.github.io | Member, Engineers Australia

PROFESSIONAL SUMMARY

Mechatronics engineer specialising in robotics, AI/ML and end-to-end automation across mechanical design, electronics, embedded firmware, controls, software, data and system validation. Delivers integrated physical systems from requirements and prototyping through commissioning, verification and handover. Experience spans ROS 2 mobile robotics, embedded and IoT platforms, GMP smart-factory integration, automotive testing and advanced manufacturing. Combines hands-on fault finding with Python/C/C++, CAD, sensor fusion and evidence-driven technical documentation.

CORE TECHNICAL SKILLS

Robotics and autonomy: ROS 2 Humble, Nav2, SLAM, Gazebo, RViz, DDS, EKF/state estimation, sensor fusion, LiDAR, IMU, odometry, kinematics and PID motion control

Embedded and electronics: C/C++, ESP32, STM32, FreeRTOS, UART, I2C, SPI, CAN, ADC, PWM, PCB design, KiCad, signal conditioning, sensors, motors, drives and actuators

AI/ML and software: Python, MATLAB/Simulink, OpenCV, YOLO, data analysis, anomaly detection, TypeScript, REST APIs, Git, Linux, Docker and CI/CD

Mechanical and product: SolidWorks, Fusion 360, 3D CAD, mechanism design, prototyping, 3D printing, instrumentation, engineering drawings and manufacturing support

Automation and integration: Requirements, controls integration, PLC/HMI/SCADA, Modbus, Profinet, MQTT, LoRaWAN AU915, field devices, FAT/SAT, commissioning and root-cause analysis

Validation and delivery: Verification and validation, test automation, CANoe/CANalyzer, ADR/EURO testing, FMEA, ISO/IEC/AS/NZS, JIRA, technical documentation and stakeholder support

PROFESSIONAL EXPERIENCE

Automation Engineer

Jan 2026 - Jun 2026

JAG Process Solutions Pty Ltd | Brunswick, VIC

- Delivered end-to-end automation and systems integration for pharmaceutical, biotech and food manufacturing under GMP, spanning control logic, HMI/process visualisation, field devices, drives, production data, testing and handover.
- Executed an iFIX-to-PVI+ migration, verified functional behaviour against the validated system, and produced FAT, SAT, commissioning, qualification and handover evidence.
- Supported MiR autonomous mobile robot operations and material-flow workflows in a smart-factory environment, diagnosing issues across robot, control, network and production interfaces.

Technical Assistant

Jul 2024 - Aug 2025

ABMARC | South Geelong, VIC

- Conducted ADR/EURO vehicle emissions, EV/PHEV range and systems testing using calibrated instrumentation, data acquisition and CAN tools; analysed deviations and supported evidence-based fault isolation.
- Prepared QA, regulatory and test documentation for certification and audit, maintaining traceable configurations, results and technical findings.

Consultant Engineer, IoT and Projects Administrator

Feb 2024 - Aug 2024

DuxTel Pty Ltd | Greater Geelong, VIC

- Designed and deployed end-to-end IoT systems linking sensors, embedded devices, gateways, Linux services, data pipelines and dashboards using ESP32, LoRaWAN, MQTT/ChirpStack, InfluxDB and Grafana.
- Integrated CAN trace, GPS, custom PCB, environmental sensing and MikroTik networking for agricultural equipment monitoring; ran field trials and supported provisioning, fault finding and handover.

Thornton Engineering, IDL and Carbon Revolution | Geelong and Corio, VIC

- Supported structural-steel engineering and QA through drawing review, ITPs, MDRs, inspection planning, material traceability and AS/NZS compliance records.
- Operated and monitored beverage and carbon-fibre production equipment while maintaining safety, quality, traceability and first-level troubleshooting discipline.

SELECTED ENGINEERING PROJECTS

- **Autonomous Navigation Rover:** Developed a modular differential-drive ROS 2 Humble autonomy stack using Gazebo and RViz, Nav2 and SLAM, PID motion control, and EKF fusion of wheel odometry and IMU data for localisation and obstacle-aware navigation.
- **ESP32 Clinical Ataxia Assessment Device:** Designed and tested an ESP32-WROOM measurement prototype with four Hall-effect sensors, 100 Hz acquisition, a real-time state machine and MATLAB analysis; executed accuracy, direction, reversal, drift and temperature protocols across 17 recorded trials.
- **Agricultural Equipment Telemetry Platform:** Built a CAN/GPS monitoring workflow with a custom PCB, ESP32 sensing, LoRaWAN AU915 and MQTT telemetry, ChirpStack, MikroTik networking, Linux services, InfluxDB/Grafana dashboards and field trials.

EDUCATION

Bachelor of Mechatronics Engineering (Honours) Deakin University, Geelong, VIC Distinction	2025
Higher National Diploma, Mechatronics, Robotics and Automation Engineering Cardiff Metropolitan University, United Kingdom Distinction	2016

PROFESSIONAL DETAILS

Membership: Member, Engineers Australia | Current Victorian Driver Licence
Selected training: Lean Six Sigma Foundation | JIRA and Agile | KAIZEN | Industrial Automation and IIoT | AI/ML | CAD
Languages: English, Tamil and Sinhala

REFERENCES

Available upon request.